Graduate Research Interests

Yohannes Alelign Biresaw

Artificial Intelligence, Machine Learning, Cybersecurity, Cryptography and Secure Systems

My central research interest is the development of secure and trustworthy intelligent systems. I am interested in how machine-learning models can remain reliable when data are sensitive, distributed, imbalanced, heterogeneous, or exposed to security threats. This broad theme connects my interests in AI and machine learning, cybersecurity, applied cryptography, privacy-preserving computation, and secure distributed systems.

Artificial Intelligence and Machine Learning

I am interested in robust learning, trustworthy AI, model calibration, generalization, uncertainty, and careful evaluation. My current research portfolio uses federated medical imaging as a concrete case study, but the underlying questions apply more broadly to intelligent systems working with real-world data.

Cybersecurity and AI Security

I would like to explore both the use of machine learning for cybersecurity and the security of machine-learning systems themselves. Relevant directions include anomaly and intrusion detection, adversarial robustness, data poisoning, model integrity, privacy leakage, and secure deployment.

Cryptography and Privacy-Preserving Computing

I am also interested in applied cryptography and privacy-preserving mechanisms that support collaboration without exposing sensitive information. Secure aggregation, cryptographic protocols for distributed learning, differential privacy, and the practical trade-offs between security, privacy, communication, and model utility are especially relevant to my longer-term goals.

Current Preparation

My completed federated-learning project provided hands-on experience in patient-aware data preparation, CNN development, non-IID partitioning, multi-seed evaluation, validation-based model selection, client-drift analysis, and reproducible reporting. My professional IT support and networking experience has also strengthened my interest in systems, operational security, and dependable infrastructure.

Graduate Objective

For graduate study, I hope to join a research group where I can focus on one well-defined problem aligned with the advisor's expertise while developing stronger foundations in mathematics, algorithms, security, and experimental research. I am especially motivated by projects that connect intelligent models with privacy, robustness, secure systems, or high-stakes applications.